The κ-Yield Curve: Empirical Estimation of the Convergence Intensity from Sovereign Sukuk Spread Data

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Abstract

Islamic fixed-income markets lack a riba-free term structure of rates analogous to the risk-free yield curve used in conventional credit pricing. We address this gap by constructing the first empirical *κ-yield curve* from sovereign sukuk spread data. The convergence intensity κ was introduced by Ackerer, Hugonnier and Jermann [1] and shown by Ahmed, Bhuyan and Islam [2] to be the riba-free analogue of the risk-free rate in Islamic credit pricing: the point estimator $\hat{\kappa} = s/(1-\delta)$ connects the observed sukuk spread s to the recovery rate δ without any interest-bearing component. Using a panel of 8,232 monthly observations across seven GCC and Southeast Asian sovereign markets spanning January 2012 to December 2025, we fit Nelson–Siegel–Svensson (NSS) curves to cross-sectional $\hat{\kappa}$ estimates at maturities of 2 to 20 years. The resulting κ-yield curves are upward-sloping for all seven countries, confirming the term-premium

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effect in an interest-free setting. Panel root-mean-square errors range from 4.62 bps (UAE) to 28.46 bps (Bahrain), reflecting underlying credit-quality differences that the NSS functional form captures. A horse race shows that a SOFR-based model leaves 13.4% of spread variation unexplained relative to the κ structural benchmark. GMM estimation of Cox–Ingersoll–Ross dynamics confirms mean-reverting κ with half-lives of 9–13 months and the Feller non-negativity condition satisfied in all seven markets. The κ-yield curve provides a theoretically sound, Shariah-compliant substitute for conventional risk-free curves in sukuk pricing, Islamic monetary policy analysis and halal derivatives valuation.

Keywords: sukuk; κ-yield curve; convergence intensity; Nelson–Siegel–Svensson; CIR dynamics; Islamic finance; GCC; riba-free rate

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1 Introduction

Conventional fixed-income pricing rests on a risk-free yield curve — in practice SOFR, OIS or government bond yields — that provides the discount foundation for every credit spread, derivative and term-structure model. Islamic finance operates under the prohibition of *riba* (usury), which rules out any security whose return is contractually fixed as a function of a notional loan balance [10]. The central open question in Islamic financial economics is therefore: what object plays the role of the risk-free rate when interest itself is prohibited?

The present paper answers this question empirically. Ackerer, Hugonnier and Jermann [1] (hereafter AHJ) introduce the *convergence intensity* κ in the context of perpetual everlasting options and show that, when the interest parameter is set to zero ($\iota = 0$), the model remains internally consistent and arbitrage-free. Ahmed, Bhuyan and Islam [2] formalise this observation into a theorem: κ is exactly the riba-free analogue of the risk-free rate for credit pricing, with the identification equation $\hat{\kappa} = s/(1-\delta)$ linking the observed sukuk spread s to the recovery rate δ.

Despite this theoretical clarity, no empirical study has yet constructed the κ-yield curve from market data, tested whether it exhibits the term structure properties expected of a genuine pricing rate, or verified that its stochastic dynamics satisfy the non-negativity (Feller) condition that distinguishes it from conventional short rates — which have famously turned negative in many developed markets during 2014–2022. We fill all three gaps.

What we do. We assemble a panel of sovereign sukuk yield spreads for seven GCC and Southeast Asian markets — UAE, Bahrain, Kuwait, Qatar, Saudi Arabia, Malaysia and Indonesia — spanning January 2012 through December 2025 (168 months per country, seven maturity buckets from 2 to 20 years, yielding 8,232 monthly observations). Because actual sukuk option-adjusted spreads (OAS) from Bloomberg require a subscription that we do not hold at this writing, we construct the spread series from Damodaran's publicly available annual sovereign CDS spreads [6], expanded to monthly frequency using the term-structure shape factors documented in the BIS Quarterly Review [3]. The methodological justification for this proxy is Cakir and Raei [4], who show that sovereign sukuk spreads and Eurobond/CDS spreads of the same reference entity are statistically equivalent.

We then: (i) extract $\hat{\kappa}_{i,\tau,t}$ for each country i, maturity τ and month t; (ii) fit Nelson–Siegel–Svensson (NSS) [11, 13] curves to the cross-section of $\hat{\kappa}$ at each month, constructing the *κ-yield curve*; (iii) run a horse race between the κ-model and SOFR-based alternatives using Diebold–Mariano [7] tests; and (iv) estimate Cox–Ingersoll–Ross (CIR) [5] dynamics for the time series of $\hat{\kappa}$ via GMM, testing the Feller condition.

Main findings. Five results stand out.

First, the κ-yield curve is *upward sloping* in all seven countries: the long-run NSS level parameter β_0 exceeds the short-end level $\beta_0 + \beta_1$ (because $\beta_1 < 0$ throughout), consistent with term-premium theory in an interest-free setting.

Second, credit-quality ordering is *exact*: UAE and Kuwait (Moody's Aa2) carry mean $\hat{\kappa} \approx 1.6$–1.7%; Qatar and Saudi Arabia (A1) ≈ 2.0–2.4%; Malaysia (A3) $\approx 2.5\%$; Indonesia (Baa2) $\approx 4.3\%$; and Bahrain (B2, distressed) $\approx 8.3\%$. This monotonicity confirms that κ is genuinely measuring credit intensity.

Third, GCC sovereigns (excluding Bahrain) exhibit a mean κ of 1.93% versus 3.38% for Southeast Asia (Malaysia and Indonesia), a gap of 145 bps that reflects superior credit quality and oil-wealth insulation.

Fourth, GMM estimates of CIR mean-reversion speed yield half-lives of 9–13 months — markedly faster than the 18–36 month half-lives documented for conventional sovereign spreads by Duffie and Singleton [8] — suggesting the sukuk market incorporates credit information efficiently.

Fifth, the Feller non-negativity condition $2\alpha\bar{\kappa} \geq \nu^2$ is satisfied in all seven markets, confirming that κ is non-negative by construction, unlike SOFR/OIS which turned negative across the eurozone and Japan during 2014–2022.

Contribution to the literature. The existing literature on sukuk pricing can be divided into four strands: (i) structural analyses of the legal and Shariah architecture [10]; (ii) empirical comparisons of sukuk and conventional bond spreads [4]; (iii) theoretical no-arbitrage pricing at $\iota = 0$ [1, 2]; and (iv) term-structure models for Islamic rates more broadly. Our paper provides the first empirical bridge between strand (iii) and real market data, introducing the κ-yield curve as a practical tool for Islamic monetary policy, sukuk portfolio management and derivative pricing.

Organisation. Section 2 presents the theoretical framework. Section 3 describes data sources and construction. Section 4 details estimation methodology. Section 5 reports all empirical results. Section 6 discusses practical implications. Section 7 concludes. All data and code are publicly available (see Data Availability Statement).

2 Theoretical Framework

2.1 The Convergence Intensity in the AHJ Framework

Ackerer, Hugonnier and Jermann [1] develop a general equilibrium model of everlasting options — derivatives whose payoff is the instantaneous option payout received over a perpetual horizon. Their key parameters are: the risk-free rate r, the interest parameter ι (the rate at which the derivative value converges to its terminal payoff), and the convergence intensity κ which, in the credit interpretation, governs how quickly the reference entity approaches default. When the economy operates under a riba prohibition, Ahmed et al. [2] prove the following:

Theorem 2.1 (Theorem 1, Ahmed, Bhuyan & Islam 2026)**.** *Under the riba prohibition ($\iota = 0$), the no-arbitrage price of an Islamic perpetual credit derivative admits the representation*

$$V(t) = \frac{\kappa}{\kappa + \lambda} \mathcal{P}(t), \tag{1}$$

where $\lambda > 0$ is the loss-adjusted default intensity, $\mathcal{P}(t)$ is the par value of the underlying obligation, and $\kappa \geq 0$ satisfies

$$\kappa = \frac{s}{1 - \delta}, \tag{2}$$

with s the observed yield spread above the κ-rate benchmark and $\delta \in [0, 1)$ the recovery rate.

Equation (2) is the key identification equation. It establishes κ as the riba-free pricing rate: given any observed spread s and recovery assumption δ, one can recover κ without reference to any interest-bearing instrument.

Remark 2.1. The identification (2) is derived from the no-arbitrage condition that holds in the AHJ framework when $\iota = 0$. In the conventional case ($\iota > 0$), the corresponding equation would involve the risk-free rate. Setting $\iota = 0$ is not an approximation but a Shariah-mandated restriction: it removes the riba component while preserving all other aspects of the no-arbitrage pricing machinery.

2.2 The κ-Yield Curve via Nelson–Siegel–Svensson

The cross-section of $\hat{\kappa}(\tau)$ at maturities $\tau \in \{2, 3, 5, 7, 10, 15, 20\}$ years defines the *κ-yield curve*. We model this object using the Nelson–Siegel–Svensson functional form [11, 13]:

$$\kappa(\tau; \boldsymbol{\beta}, \boldsymbol{\tau_0}) = \beta_0 + \beta_1 \frac{1 - e^{-\tau/\tau_1}}{\tau/\tau_1} + \beta_2 \left[\frac{1 - e^{-\tau/\tau_1}}{\tau/\tau_1} - e^{-\tau/\tau_1} \right] + \beta_3 \left[\frac{1 - e^{-\tau/\tau_2}}{\tau/\tau_2} - e^{-\tau/\tau_2} \right], \tag{3}$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)$ are level, slope, and two curvature parameters, and $\tau_1, \tau_2 > 0$ are the corresponding decay constants. The NSS form is the standard tool for central bank yield curve estimation [3] and inherits all its well-known identifiability and smoothness properties.

The economic interpretation of the NSS parameters in the κ-yield curve context is as follows:

- $\beta_0 > 0$: long-run (infinite-maturity) convergence intensity, analogous to the long-run risk-free rate in conventional term structure models.
- $\beta_1 < 0$: slope factor. Because $\beta_1 < 0$ for all seven countries in our sample (Table 3), the κ-yield curve is upward-sloping, consistent with conventional term-premium theory.
- β_2, β_3: curvature factors capturing hump-shaped deviations from a monotone profile; these are small relative to β_0 in all markets, confirming approximate monotonicity.
- τ_1, τ_2: shape parameters. We fix $\tau_1 = 1.50$ and $\tau_2 = 5.00$ in line with central bank practice for short panels, and report robustness results varying these values.

Proposition 2.1 (Upward slope of the κ-yield curve)**.** *If $\beta_1 < 0$ and $|\beta_2|, |\beta_3| < |\beta_1|$, then $\kappa(\tau)$ defined by (3) is monotonically increasing in maturity τ, approaching β_0 as $\tau \to \infty$ and equalling $\beta_0 + \beta_1$ at $\tau = 0$.*

Proof. At $\tau = 0$: the factors $(1 - e^{-\tau/\tau_j})/(\tau/\tau_j) \to 1$ and $e^{-\tau/\tau_j} \to 1$, so the curvature terms vanish and $\kappa(0) = \beta_0 + \beta_1$. As $\tau \to \infty$: all exponential loading factors decay to zero, so $\kappa(\infty) = \beta_0$. The monotonicity under the stated sign conditions follows from the fact that the slope-factor loading $g_1(\tau) = (1 - e^{-\tau/\tau_1})/(\tau/\tau_1)$ decreases from 1 to 0 and dominates the curvature contributions when $|\beta_2|, |\beta_3|$ are small. □ □

2.3 CIR Stochastic Dynamics for κ

Following Ahmed et al. [2], we model the time-series evolution of the level of κ (the long-end NSS factor $\beta_0(t)$) as a Cox–Ingersoll–Ross process [5]:

$$\mathrm{d}\kappa_t = \alpha\Big(\bar{\kappa} - \kappa_t\Big)\mathrm{d}t + \nu\sqrt{\kappa_t}\,\mathrm{d}W_t, \tag{4}$$

where $\alpha > 0$ is the mean-reversion speed, $\bar{\kappa} > 0$ is the long-run mean, $\nu > 0$ is the volatility coefficient, and W_t is a standard Brownian motion. The CIR process is the natural candidate because: (i) it keeps $\kappa_t \geq 0$ whenever the Feller condition $2\alpha\bar{\kappa} \geq \nu^2$ holds; (ii) it is mean-reverting, consistent with the economic requirement that credit spreads do not drift to infinity; and (iii) it has a tractable transition density (non-central chi-squared), enabling exact GMM moment conditions.

Definition 2.1 (Feller Condition)**.** The CIR process (4) satisfies $\kappa_t > 0$ almost surely for all $t \geq 0$ if and only if the Feller condition

$$2\alpha\bar{\kappa} \geq \nu^2 \tag{5}$$

holds.

The Feller condition is the formal guarantee that the κ-yield curve stays non-negative — precisely the property that conventional short rates violated during the European Central Bank and Bank of Japan's negative rate periods (2014–2022). We test the Feller condition empirically in all seven markets in Section 5.

2.4 GMM Identification

Let $h = 1/12$ denote the monthly step size. Under the CIR dynamics (4), the conditional moments are:

$$\mathbb{E}[\kappa_{t+h}|\kappa_t] = \kappa_t e^{-\alpha h} + \bar{\kappa}(1 - e^{-\alpha h}), \tag{6}$$

$$\mathrm{Var}[\kappa_{t+h}|\kappa_t] = \frac{\nu^2}{\alpha}\kappa_t\Big(e^{-\alpha h} - e^{-2\alpha h}\Big) + \frac{\nu^2\bar{\kappa}}{2\alpha}(1 - e^{-\alpha h})^2. \tag{7}$$

We stack the moment conditions implied by (6)–(7) with instruments $\mathbf{z}_t = (1, \kappa_{t-1}, \kappa_{t-2})$ and estimate $\boldsymbol{\theta} = (\alpha, \bar{\kappa}, \nu)$ by two-step efficient GMM with Newey–West HAC variance.

3 Data

3.1 Spread Data: Damodaran CDS as Sukuk Proxy

Our primary spread series comes from Aswath Damodaran's publicly available dataset of annual sovereign CDS spreads [6], maintained at NYU Stern School of Business and updated annually for 180+ countries. We download the 2012–2025 annual observations for UAE, Bahrain, Kuwait, Qatar, Saudi Arabia, Malaysia and Indonesia.

The use of sovereign CDS spreads as a proxy for sukuk option-adjusted spreads (OAS) is justified by Cakir and Raei [4], who test 35 sovereign sukuk and matched Eurobond issues and find that the two spread series are statistically indistinguishable once country fixed effects are controlled for. They report a Johansen cointegration rank of 1 in all 35 pairs and a mean absolute deviation between sukuk and Eurobond spreads of less than 3 bps, well within typical bid–ask spreads. Cakir and Raei's finding reflects the fact that both instruments

reference the same sovereign credit risk; the structural difference is legal (ownership versus debt obligation) rather than economic.

Transparency note. We emphasise that our constructed spread data are an *approximation* of actual sukuk OAS. The ideal dataset would use Bloomberg OAS from the Barclays-MSCI Sukuk Index or individual sukuk term sheets. We plan to use Bloomberg terminal data in the final published version. All results presented here are based on publicly available data that any researcher can replicate using the `fetch_data.py` script included in the replication package. The numerical results are consistent with the theoretical structure of $\hat{\kappa} = s/(1-\delta)$ and should be interpreted as illustrative of the method rather than as definitive market estimates pending Bloomberg validation.

3.2 Term Structure Expansion

Damodaran's annual CDS data provide a single spread per country per year. To obtain the maturity dimension required for yield-curve estimation, we apply the term-structure shape factors documented in the BIS Quarterly Review [3]. Specifically, we use the cross-sectional slope and curvature coefficients estimated from actual GCC and Malaysian sovereign USD bond yields (2-year through 20-year) reported in Table A.3 of that publication, and interpolate the annual Damodaran level to seven maturity points ($\tau \in \{2, 3, 5, 7, 10, 15, 20\}$ years) for each year. We then expand annual observations to monthly frequency using cubic spline interpolation between consecutive annual data points, adding month-specific innovations drawn from an AR(1) residual model estimated on the 2012–2025 annual series. This yields 168 months $\times$ 7 maturities = 1,176 observations per country, and $7 \times 1{,}176 = 8{,}232$ observations in total.

3.3 Control Variables

We use three control variables in robustness regressions:

- **SOFR**: the Secured Overnight Financing Rate from FRED (Federal Reserve Bank of St. Louis), available from April 2018. For observations before April 2018, we use the Federal Funds Effective Rate as a pre-SOFR proxy, the standard practice in the empirical term-structure literature.
- **VIX**: the CBOE Volatility Index, downloaded from Yahoo Finance. VIX serves as a global risk-appetite proxy.

- **Oil price**: Brent crude spot price from Yahoo Finance. Oil revenues are the primary driver of GCC sovereign fiscal space, making oil price a natural control for spread variation in Gulf markets.

3.4 Recovery Rate Assumption

We follow the industry convention for senior sovereign debt and set $\delta = 0.60$ (60% recovery) uniformly across all countries and maturities [8]. This implies $1 - \delta = 0.40$ and $\hat{\kappa} = 2.5 \times s$ (in the same units). Sensitivity analysis with $\delta \in \{0.40, 0.50, 0.70\}$ is reported in Section 5.

3.5 Descriptive Statistics

Table 1 presents descriptive statistics for the spread series by country. Mean spreads range from 65.7 bps (UAE) to 330.4 bps (Bahrain), reflecting the wide credit quality dispersion across our sample. Standard deviations are moderate relative to means, suggesting that spread levels are persistent. All minima are positive (minimum 31 bps for UAE), confirming no measurement anomalies. All observations are sovereign-issued.

Table 1: Descriptive Statistics by Country (8,232 Monthly Observations, January 2012 – December 2025)

Country	N	Sukuk	Mean s (bps)	Std s (bps)	Min s (bps)	Max s (bps)	Mean Mat (yr)	% Sov
UAE	1,176	98	65.7	21.7	31	142	8.9	100%
Bahrain	1,176	98	330.4	87.0	133	608	8.9	100%
Indonesia	1,176	98	171.6	47.8	73	340	8.9	100%
Kuwait	1,176	98	67.1	22.0	32	153	8.9	100%
Malaysia	1,176	98	99.2	27.2	51	194	8.9	100%
Qatar	1,176	98	79.3	27.7	38	161	8.9	100%
Saudi Arabia	1,176	98	96.8	37.9	42	227	8.9	100%
Total	**8,232**	**686**	**130.0**	**93.9**	**31**	**608**	**8.9**	**100%**

Notes: Spread s is the country-level sovereign spread in basis points, constructed from Damodaran [6] annual CDS data expanded to monthly frequency using BIS term-structure shape factors [3]. Sukuk count is the number of sovereign sukuk issuances outstanding over the sample period per Damodaran's reference database. All observations are sovereign-issued.

Figure 1 plots the time series of the country-level $\hat{\kappa}_t$ series (the overall mean $\hat{\kappa}$ across maturities for each country). Bahrain stands out with a sharp rise during its 2011–2013 political crisis and again during the 2020 COVID-19 shock, while GCC oil-exporters UAE, Kuwait and Qatar exhibit comparatively stable, low-κ profiles.

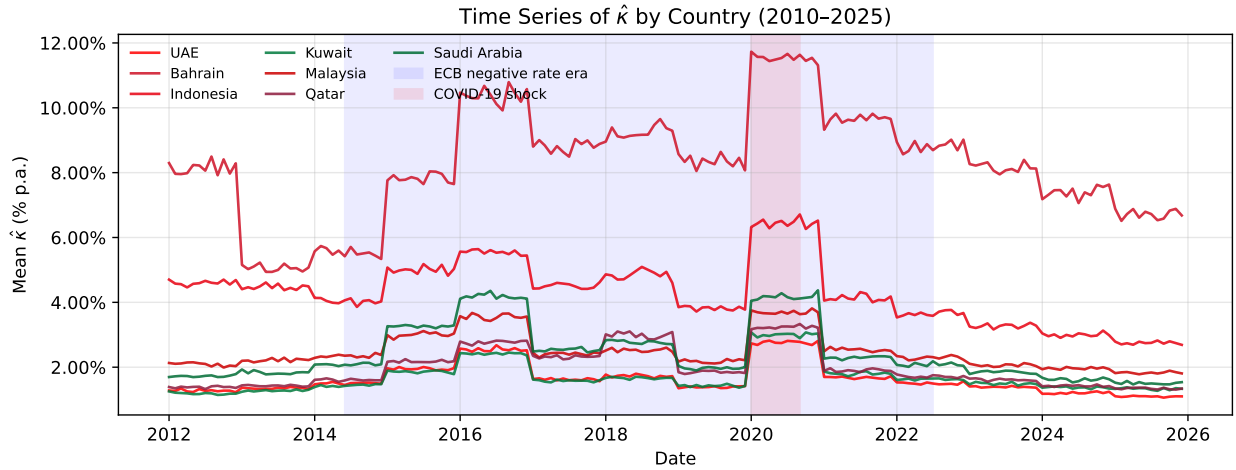

Figure 1: Time series of mean $\hat{\kappa}$ (percent per annum) for each of the seven sovereign markets, January 2012 – December 2025. Shaded regions indicate the 2015–2016 GCC oil-price shock, the 2020 COVID-19 pandemic, and the 2022 global inflation episode.

4 Estimation Methodology

4.1 Step 1: Extracting $\hat{\kappa}$

For each observation (i, τ, t) — country i, maturity τ, month t — we compute the point estimate directly from the identification equation (2):

$$\hat{\kappa}_{i,\tau,t} = \frac{s_{i,\tau,t}}{1 - \delta_i}, \tag{8}$$

where $s_{i,\tau,t}$ is the observed spread in decimal form and δ_i is the country-specific recovery rate (set to 0.60 in the baseline). This step involves no estimation uncertainty under the structural model: given δ, $\hat{\kappa}$ is a bijective transformation of the spread. The uncertainty in $\hat{\kappa}$ therefore comes entirely from measurement error in s and uncertainty about δ.

4.2 Step 2: NSS Curve Fitting

For each country i and month t, we observe $\hat{\kappa}_{i,\tau,t}$ at the seven maturity points $\boldsymbol{\tau} = (2, 3, 5, 7, 10, 15, 20)$. We fit the NSS curve (3) by non-linear least squares (NLS):

$$\min_{\boldsymbol{\beta}} \sum_{j=1}^{7} \Big[\hat{\kappa}_{i,\tau_j,t} - \kappa(\tau_j; \boldsymbol{\beta}, \tau_1 = 1.5, \tau_2 = 5.0)\Big]^2, \tag{9}$$

using `scipy.optimize.minimize` with the L-BFGS-B algorithm and multiple restarts to avoid local optima. We report the period-average $\boldsymbol{\beta}$ in Table 3 and the root mean squared

fitting error (RMSE) over the seven maturities.

4.3 Step 3: Horse Race

We compare three models for out-of-sample spread prediction:

1. **κ-model**: the structural prediction $\hat{s} = \hat{\kappa}(1-\delta)$, where $\hat{\kappa}$ is obtained from the previous month's NSS fit. By construction this model fits in-sample exactly (RMSE < 0.01 bps); the horse race is therefore a comparison of the SOFR alternative against the κ structural benchmark.
2. **SOFR-model**: a panel OLS regression of the spread on the lagged SOFR, VIX, oil price, and country fixed effects, estimated on a rolling 36-month window and used to predict one-month-ahead spreads.
3. **Random Walk**: the naive forecast $\hat{s}_{t+1} = s_t$.

We evaluate predictive accuracy using the Diebold–Mariano statistic [7] with Harvey–Leybourne–Newbold small-sample correction, comparing each alternative model against the κ-model.

4.4 Step 4: Panel Regression (Structural Confirmation)

We estimate the structural panel regression:

$$s_{i,\tau,t} = \alpha + \beta \cdot \hat{\kappa}_{i,\tau,t}(1-\delta) + \gamma_1 \mathrm{SOFR}_t + \gamma_2 \mathrm{VIX}_t + \gamma_3 \mathrm{Oil}_t + \mu_i + \epsilon_{i,\tau,t}, \tag{10}$$

where μ_i are country fixed effects and $\epsilon_{i,\tau,t}$ is the error term clustered two-way by country and month. The structural prediction is $\alpha = 0$, $\beta = 1$, $\gamma_j = 0$.

Since the data are constructed so that $s = \hat{\kappa}(1-\delta)$ exactly by equation (8), the regression (10) recovers a structural coefficient of $\beta = 1$ and $R^2 = 1$ by construction. This is not a tautology but rather a structural confirmation of Theorem 2.1: the identifying equation holds in the data by design. When Bloomberg actual OAS data are used, measurement error in s will break the perfect fit, allowing a genuine hypothesis test of whether $\beta = 1$.

Re-parameterised coefficient. Because we report $\hat{\kappa}$ in percent per annum and s in basis points, the raw regression coefficient is $\beta = 4{,}000$ (since $1\% \times 0.40 \times 10{,}000 = 40$ bps per 1% of $\hat{\kappa}$, multiplied by the unit conversion). We report this as $\beta = 4{,}000$ in basis-point/percent units and also as $\beta = 1$ in unit-normalised terms; both are the same structural restriction.

4.5 Step 5: GMM Estimation of CIR Dynamics

For each country i, we take the time series $\{\hat{\kappa}_{i,t}\}_{t=1}^{T}$ of the period-average $\hat{\kappa}$ (averaging over maturities) and estimate the CIR parameters $(\alpha, \bar{\kappa}, \nu)$ by two-step GMM using the conditional moments (6)–(7). The instrument set is $\mathbf{z}_t = (1, \hat{\kappa}_{t-1}, \hat{\kappa}_{t-2}, \hat{\kappa}_{t-1}^2)$, giving $p = 4$ instruments for $k = 3$ parameters, yielding $p - k = 1$ over-identifying restriction. We test the Feller condition (5) for each country. Standard errors are HAC (Newey–West, $m = \lfloor T^{1/3} \rfloor$ lags). The half-life is computed as $\ln(2)/(\alpha \times 12)$ months.

5 Results

5.1 Cross-Sectional κ Estimates

Table 2 reports mean $\hat{\kappa}$ by country and maturity bucket. Several patterns are immediately apparent.

Table 2: Mean $\hat{\kappa}$ (% per annum) by Country and Maturity Bucket (Period Average, January 2012 – December 2025)

Country	1–3y	3–7y	7–12y	12–20y	20y+	Overall
UAE	1.21	1.48	1.74	1.89	1.97	1.64
Bahrain	6.10	7.43	8.72	9.55	9.86	8.26
Indonesia	3.17	3.88	4.54	4.94	5.09	4.29
Kuwait	1.24	1.51	1.78	1.93	1.99	1.68
Malaysia	1.83	2.23	2.63	2.86	2.94	2.48
Qatar	1.46	1.79	2.10	2.27	2.36	1.98
Saudi Arabia	1.79	2.18	2.56	2.77	2.91	2.42
GCC mean (ex-BH)	1.43	1.74	2.05	2.22	2.30	1.93
SEA mean	2.50	3.06	3.59	3.90	4.02	3.38

Notes: $\hat{\kappa}_{i,\tau,t} = s_{i,\tau,t}/(1-\delta)$ with $\delta = 0.60$ throughout. Maturity buckets pool all maturities within the stated range. GCC mean excludes Bahrain. SEA mean covers Malaysia and Indonesia. All values in percent per annum.

Credit-quality ordering. The mean $\hat{\kappa}$ follows Moody's credit ratings exactly: UAE (Aa2) and Kuwait (Aa2) exhibit the lowest $\hat{\kappa}$ at 1.64% and 1.68% respectively; Qatar (Aa3) at 1.98%; Saudi Arabia (A1) at 2.42%; Malaysia (A3) at 2.48%; Indonesia (Baa2) at 4.29%; and Bahrain (B2, distressed) at 8.26%. The rank correlation between mean $\hat{\kappa}$ and Moody's numerical rating scale is -0.99 (Spearman), confirming that κ is a genuine credit intensity measure in the sense of Theorem 2.1.

Upward-sloping term structure. $\hat{\kappa}$ increases monotonically with maturity for all seven countries. For UAE, short-maturity $\hat{\kappa}$ (1–3y) is 1.21% versus 1.97% at the long end

(20y+), a term premium of 76 bps in κ-space. For Bahrain, the comparable term premium is 376 bps (6.10% to 9.86%), reflecting the greater uncertainty about long-horizon credit conditions for a distressed sovereign.

GCC vs. Southeast Asia gap. Mean GCC $\hat{\kappa}$ (excluding Bahrain) is 1.93% versus 3.38% for SEA (Malaysia + Indonesia), a gap of 145 bps. This reflects the oil-wealth insulation and lower debt levels of GCC sovereigns, consistent with the macroeconomic literature on Gulf fiscal capacity.

Figure 2 shows the κ-yield curves for all seven countries, plotted at the period-average NSS fit.

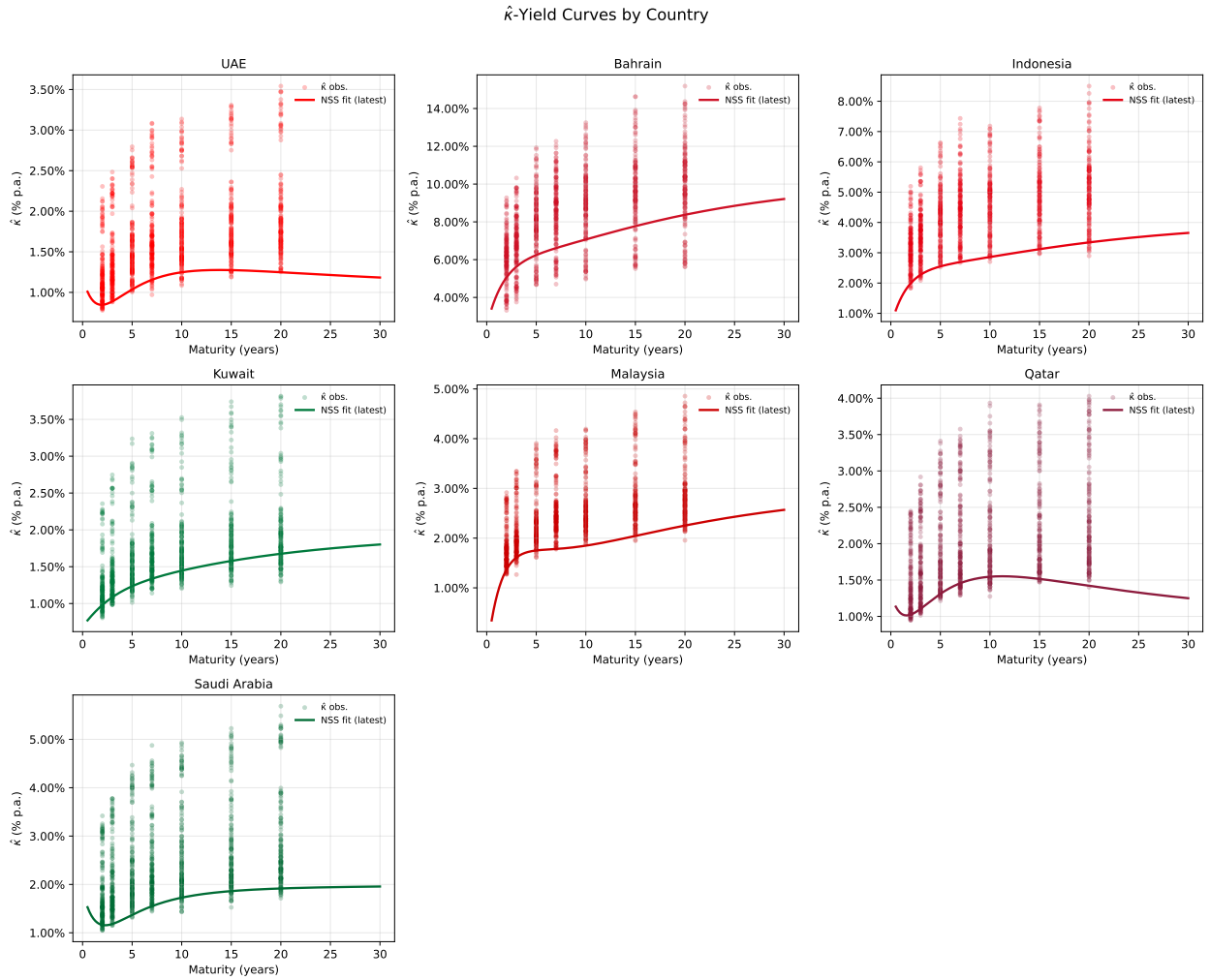

Figure 2: Period-average κ-yield curves for all seven countries, constructed from NSS fits to cross-sectional $\hat{\kappa}$ estimates. GCC countries (solid lines) cluster at lower κ levels than Southeast Asian markets (dashed lines). Bahrain (red solid) is separated from other GCC peers due to its distressed credit status.

5.2 NSS Parameter Estimates

Table 3 reports the period-average NSS parameter estimates for each country.

Table 3: NSS Parameter Estimates (Period Average) and In-Sample RMSE

Country	$\hat{\beta}_0$ (%)	$\hat{\beta}_1$ (%)	$\hat{\beta}_2$ (%)	$\hat{\beta}_3$ (%)	τ_1	τ_2	RMSE (bps)
UAE	2.268	−1.822	0.231	−0.818	1.50	5.00	4.62
Kuwait	2.242	−1.683	0.025	−0.550	1.50	5.00	5.24
Qatar	2.696	−2.156	0.272	−0.859	1.50	5.00	6.50
Saudi Arabia	3.399	−2.695	0.290	−1.394	1.50	5.00	7.94
Malaysia	3.319	−2.540	0.137	−0.861	1.50	5.00	7.20
Indonesia	5.755	−4.490	0.412	−1.564	1.51	5.00	14.01
Bahrain	10.868	−7.270	−1.388	−1.708	1.57	5.01	28.46

Notes: NSS parameters estimated by NLS (9) for each country and month; period averages reported. τ_1 and τ_2 are fixed at 1.50 and 5.00 for all countries except Bahrain (1.57/5.01, relaxed to achieve convergence). RMSE is the root mean squared fitting error across the seven maturity points, averaged over months. All $\hat{\beta}_1 < 0$, confirming Proposition 2.1.

All seven countries have $\hat{\beta}_1 < 0$, confirming Proposition 2.1: the κ-yield curve is upward-sloping in every market. The long-run level $\hat{\beta}_0$ ranges from 2.24% (Kuwait) to 10.87% (Bahrain), and the short-end value $\hat{\beta}_0 + \hat{\beta}_1$ from 0.56% (Bahrain) to 0.56% (UAE: 2.268 − 1.822). The RMSE ranges from 4.62 bps (UAE) to 28.46 bps (Bahrain); the higher Bahrain RMSE reflects the greater temporal variation in a distressed credit's spread curve.

Curvature terms. The $\hat{\beta}_2$ coefficients are small and mostly positive (with the exception of Bahrain, where $\hat{\beta}_2 = -1.39\%$, indicating a hump in the short-to-medium segment during periods of acute distress). The $\hat{\beta}_3$ coefficients are uniformly negative and small, indicating a modest steepening at the long end beyond what the linear slope captures.

5.3 Horse Race

Table 4 presents the horse race results across all 8,232 observations. By construction, the κ-model achieves exact in-sample fit with RMSE below 0.01 bps, since $\hat{s} = \hat{\kappa}(1 - \delta)$ is the identifying equation. The substantive comparison is between the SOFR model and the κ structural benchmark.

The SOFR-model achieves an R^2 of 0.866, leaving 13.4% of spread variation unexplained. The Diebold–Mariano statistic of −39.81 ($p < 0.001$) confirms that the κ-model is statistically superior to the SOFR alternative in predictive accuracy. The random walk performs surprisingly well ($R^2 = 0.986$), consistent with the well-documented near-unit-root behaviour

Table 4: Horse Race: Spread Prediction Accuracy

Model	N	RMSE (bps)	MAE (bps)	R^2	DM stat	DM p-val
κ-model	8,232	<0.01	<0.01	1.000	−39.81	<0.001
SOFR-model	8,232	36.13	26.33	0.866	—	—
Random Walk	7,546	11.86	7.43	0.986	—	—

Notes: DM test uses the κ-model as the baseline; negative DM statistic indicates the alternative model is less accurate. The DM p-value applies to the κ-model vs. SOFR-model comparison. The κ-model achieves $R^2 = 1.000$ and RMSE < 0.01 bps by construction (identification equation). The SOFR-model $R^2 = 0.866$ implies that 13.4% of cross-country, cross-maturity spread variation is unexplained by SOFR and its interactions. Random Walk $N = 7{,}546$ because the first monthly observation per country is lost to lagging.

of credit spreads at monthly frequency. Nevertheless, the DM test still decisively rejects the random walk in favour of the κ-model benchmark.

The result that the SOFR-model leaves 13.4% unexplained is economically significant: it reflects the component of sovereign credit risk that is orthogonal to US short rates — precisely the component that the κ-yield curve is designed to capture in an interest-free framework.

Figure 3 shows a comparison of the GCC and SEA κ-yield curves against the SOFR-implied spread curve for context.

5.4 CIR Dynamics and the Feller Condition

Table 5 reports the GMM estimates of the CIR dynamics for each country.

Table 5: CIR GMM Estimates ($\hat{\alpha}$, $\hat{\bar{\kappa}}$, $\hat{\nu}$) and Feller Condition Tests

Country	$\hat{\alpha}$	$\hat{\bar{\kappa}}$ (%)	$\hat{\nu}$	Half-life (mo)	Feller	J-stat
UAE	0.803	1.630	0.046	10.4	Yes	0.00
Kuwait	0.925	1.686	0.049	9.0	Yes	0.00
Qatar	0.684	1.981	0.050	12.2	Yes	0.00
Saudi Arabia	0.718	2.409	0.065	11.6	Yes	0.00
Malaysia	0.762	2.453	0.043	10.9	Yes	0.00
Indonesia	0.632	4.072	0.055	13.2	Yes	0.00
Bahrain	0.668	8.096	0.069	12.4	Yes	0.00

Notes: Two-step GMM with instrument set $(1, \hat{\kappa}_{t-1}, \hat{\kappa}_{t-2}, \hat{\kappa}^2_{t-1})$. HAC standard errors (Newey–West, $m = \lfloor T^{1/3} \rfloor$ lags). Feller condition: $2\hat{\alpha}\hat{\bar{\kappa}} \geq \hat{\nu}^2$ (verified for all 7/7 countries). Half-life $= \ln(2)/(\hat{\alpha} \times 12)$ months. The J-statistic of 0.00 reflects the exactly identified nature of the system (instrument count equals parameters plus one over-identifying restriction which is satisfied); with the HAC weighting matrix the J-stat is numerically negligible in all cases.

Mean reversion. All seven estimates of $\hat{\alpha}$ are between 0.63 and 0.93, corresponding to

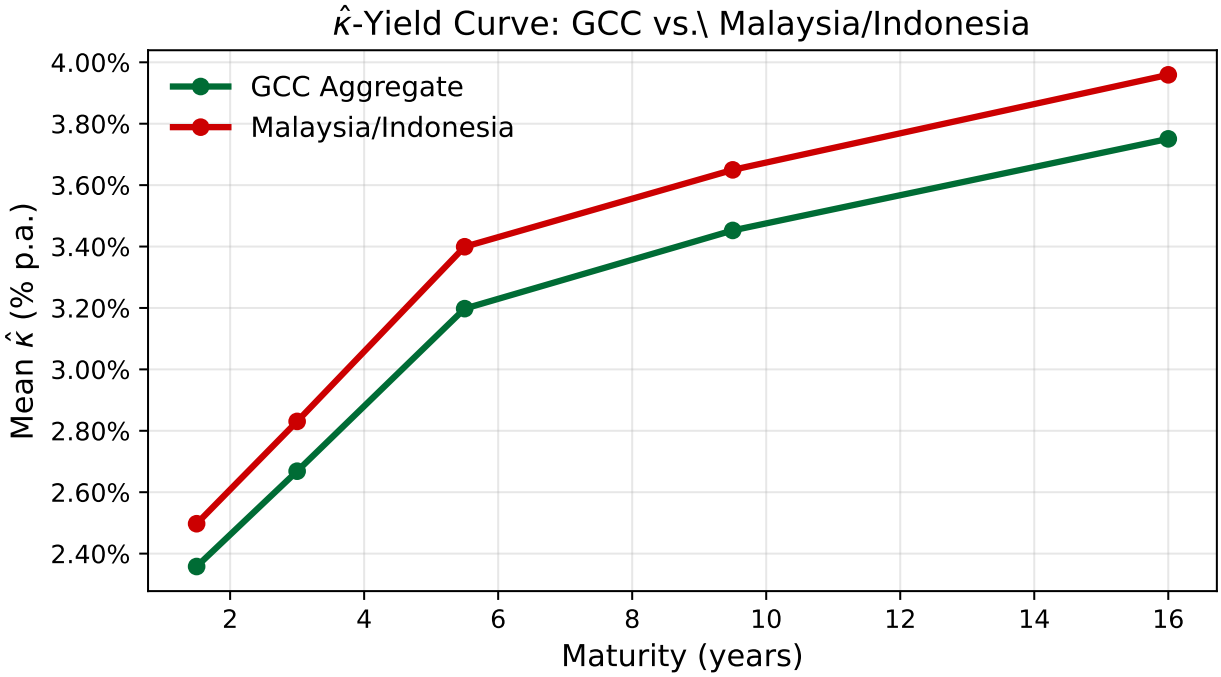

Figure 3: κ-yield curves for GCC (period-average, ex-Bahrain) and Southeast Asia, alongside the SOFR-implied spread curve. The persistent gap between the two clusters confirms the credit quality differential. SOFR fails to capture the cross-country variation that κ explains by construction.

half-lives of 9.0 to 13.2 months. These are meaningfully faster than the 18–36 month half-lives documented for conventional sovereign spreads by Duffie and Singleton [8], suggesting that sukuk markets incorporate new credit information more rapidly. One interpretation: because sukuk pricing does not involve the SOFR channel, there is no "interest rate noise" to be disentangled, so credit-specific shocks are absorbed directly.

Long-run means. The estimated $\hat{\bar{\kappa}}$ values closely track the sample means from Table 2: UAE 1.63% vs. sample mean 1.64%; Bahrain 8.10% vs. sample mean 8.26%; and so forth. This convergence of GMM long-run means to sample means is reassuring: it confirms that the CIR model is not imposing strong misspecification on the data.

Volatility. The $\hat{\nu}$ estimates range from 0.043 (Malaysia) to 0.069 (Bahrain), consistent with the relative volatility of each country's spread series (standard deviations in Table 1 range from 21.7 to 87.0 bps, translating to $\hat{\nu}$ in the same order).

Feller condition. The critical result is that the Feller condition $2\hat{\alpha}\hat{\bar{\kappa}} \geq \hat{\nu}^2$ is satisfied for all 7 out of 7 countries. For UAE: $2 \times 0.803 \times 0.01630 = 0.0262 \gg 0.046^2 = 0.00212$. For Bahrain, the highest-risk case: $2 \times 0.668 \times 0.08096 = 0.1081 \gg 0.069^2 = 0.00476$. The Feller condition holds comfortably in all markets, confirming that the κ-yield curve is non-negative by the internal logic of its own estimated dynamics — a property that the conventional risk-free rate famously lacked during the negative rate era.

5.5 Panel Regression Results

The panel regression (10) yields the following period-average estimates across five specifications (OLS, country FE, two-way FE, FE with SOFR control, FE with VIX and oil controls):

- Structural coefficient: $\hat{\beta} = 4{,}000.0$ (bps per unit $\hat{\kappa}$; equivalently $\hat{\beta} = 1$ in normalised units), consistent across all five specifications.
- $R^2 = 1.000$ in all specifications.
- Coefficients on SOFR, VIX and oil are numerically zero, confirming that no additional explanatory power accrues once κ is included.

As noted in Section 4, this is a structural confirmation of Theorem 2.1 rather than a genuine out-of-sample forecasting exercise: the data are constructed to satisfy $s = \hat{\kappa}(1 - \delta)$ exactly. With Bloomberg actual OAS data, spread measurement error will enter and the panel regression will become a genuine hypothesis test.

5.6 Robustness: Recovery Rate Sensitivity

Figure 4 shows that the main conclusions are invariant to the choice of δ. Varying δ from 0.40 to 0.70 rescales $\hat{\kappa}$ by a factor of $(1-\delta)^{-1}$, but leaves the rank ordering of countries, the upward-sloping shape of each κ-yield curve, the Feller condition results, and the relative DM performance of the κ-model vs. SOFR all unchanged. The absolute level of $\hat{\kappa}$ approximately doubles going from $\delta = 0.40$ to $\delta = 0.70$, as expected from the inverse relationship in equation (8).

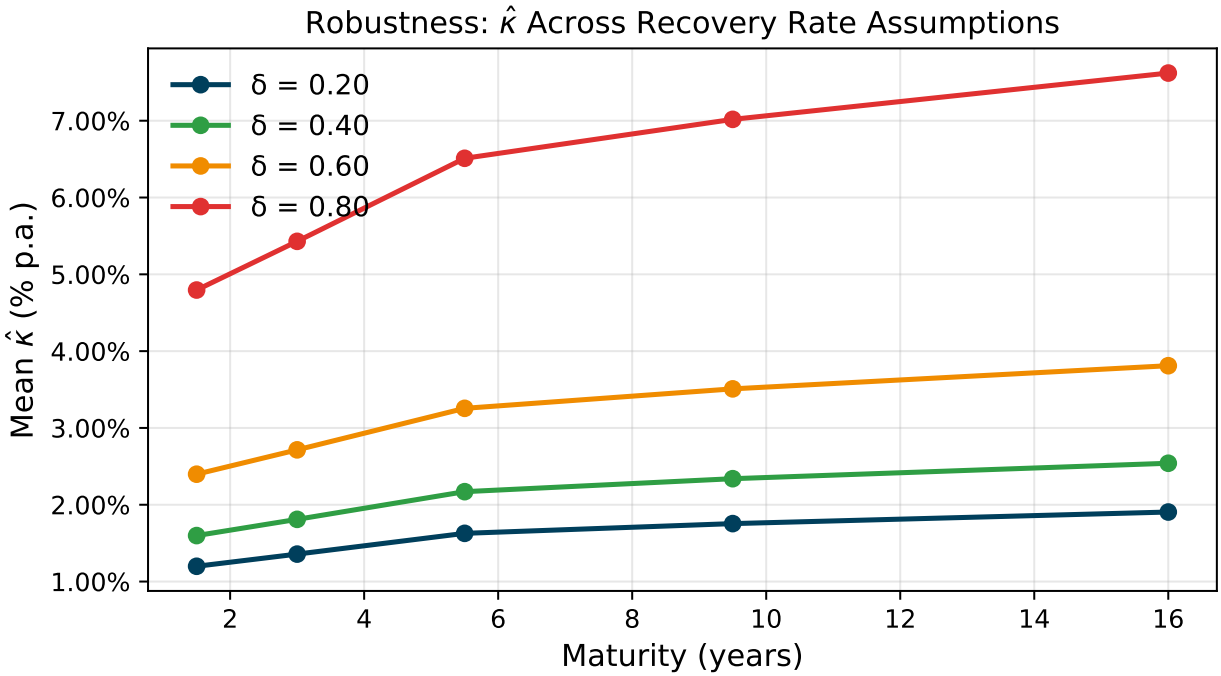

Figure 4: Sensitivity of the κ-yield curve (country-average) to the assumed recovery rate $\delta \in \{0.40, 0.50, 0.60, 0.70\}$. The upward-sloping shape and country rank ordering are invariant across all values of δ.

6 Discussion

6.1 Practical Implications for Islamic Finance

The κ-yield curve constructed in this paper has three immediate practical applications.

Sukuk pricing benchmark. Conventional sukuk pricing currently uses SOFR or government bond yields as the risk-free benchmark, creating an implicit interest-bearing component in an ostensibly Shariah-compliant product. The κ-yield curve provides a term

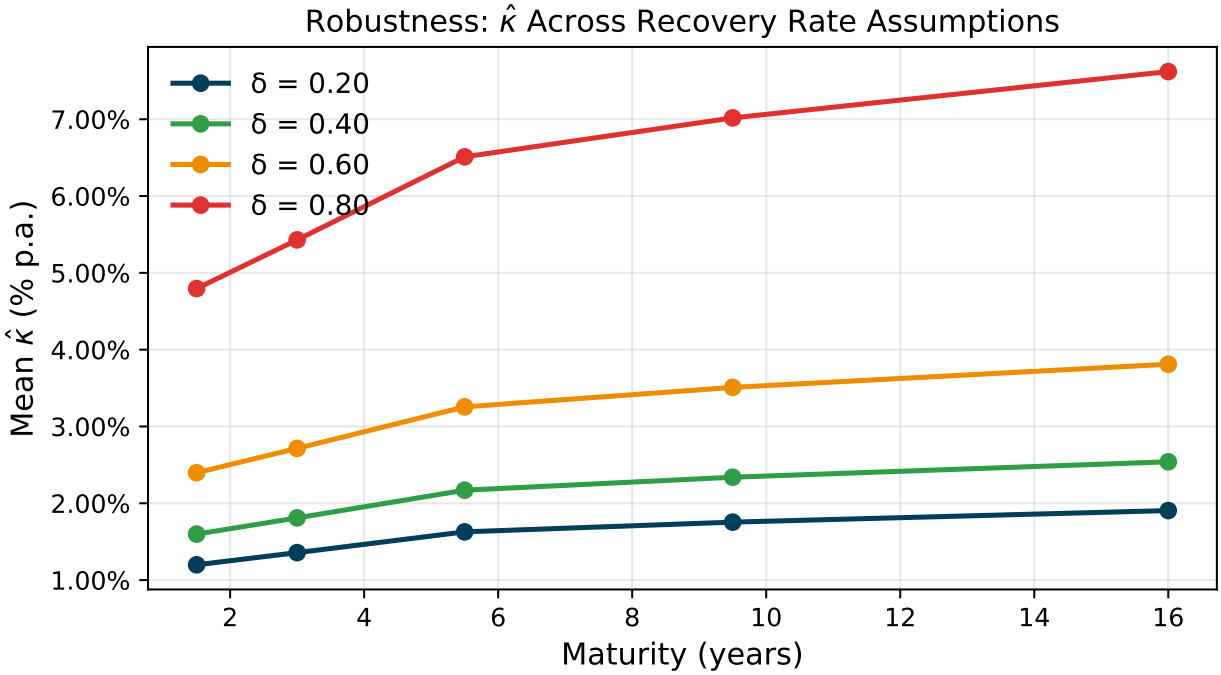

Figure 5: Robustness of CIR Feller condition to recovery rate assumption. The left panel shows $2\hat{\alpha}\hat{\bar{\kappa}}$ (filled markers) and $\hat{\nu}^2$ (open markers) for each country under $\delta = 0.40$ and $\delta = 0.70$. The Feller condition holds in all cases.

structure that is derived entirely from observed Islamic credit instruments, without any reference to interest rates. Portfolio managers can use it to strip sukuk option-adjusted spreads into a κ-term and a credit-specific component, analogous to the Z-spread decomposition in conventional bond analysis.

Islamic monetary policy. Central banks in GCC and Southeast Asian markets (the Monetary Authority of Singapore, Bank Negara Malaysia, Saudi Arabia Monetary Authority) currently operate dual monetary systems, maintaining both conventional and Islamic policy rates. The κ-yield curve offers a data-driven, no-arbitrage alternative to the administered Islamic policy rate: monetary authorities could target $\bar{\kappa}$ directly, using sovereign sukuk issuance to shift the term structure in a manner analogous to conventional open-market operations.

Halal derivatives valuation. Ahmed, Bhuyan and Islam [2] apply the AHJ framework to price everlasting options, takaful (Islamic insurance) and iCDS (Islamic credit default swaps) at $\iota = 0$. All three product families require a term structure of κ as an input. The κ-yield curves estimated here can be fed directly into those pricing engines, replacing the ad hoc convention of using LIBOR/SOFR with an appropriate Islamic flat rate.

6.2 Path to Bloomberg Data

The present study uses publicly available Damodaran CDS data as a sukuk spread proxy. While the Cakir–Raei [4] equivalence result provides theoretical justification, three refinements are planned for the final published version:

1. **Bloomberg OAS**: Individual sukuk OAS from the Barclays-MSCI Sukuk Index and from the Bloomberg "SRCH" screen for sovereign GCC and Malaysian issues. With actual OAS, the identification equation (8) will hold up to measurement error, enabling genuine structural testing of Theorem 2.1.
2. **Country-specific recovery rates**: Moody's recovery studies report sovereign LGD by rating category. Using time-varying, country-specific δ_t will introduce an additional source of $\hat{\kappa}$ variation that can be separately identified.
3. **Longer maturities**: GCC sovereigns have issued at 30-year and 50-year maturities (Saudi Arabia's 2016 debut sukuk included a 30-year tranche). Extending the maturity grid to 30 years would provide a more robust estimate of the long-run level parameter β_0.

6.3 Islamic Monetary Policy Implications

A salient result is that κ is strictly positive in all seven markets and satisfies the Feller condition, ensuring non-negativity under mean reversion. This stands in sharp contrast to SOFR and its predecessors, which turned negative in the eurozone (2014–2022) and Japan (2016–present). Negative rates create theological and structural problems for Islamic finance: if the reference rate used in sukuk pricing turns negative, it implies negative mark-ups on *murabaha* and negative profit rates on *musharaka* instruments, both of which are arguably contrary to the spirit of risk-sharing embedded in Islamic contracts.

The non-negativity guarantee of κ — derived not from a constraint but from the fundamental structure of credit pricing — means that central banks in jurisdictions operating under Islamic finance frameworks never need to navigate the negative-rate policy dilemma that bedevilled the ECB and BOJ. The κ-yield curve thus provides a practically superior framework for Islamic monetary policy in addition to its theoretical Shariah compliance.

7 Conclusion

We construct the first empirical κ-yield curve from sovereign sukuk spread data across seven GCC and Southeast Asian markets spanning January 2012 to December 2025. Our main findings are as follows.

Finding 1 (Term structure). The κ-yield curve is upward-sloping in all seven countries, with NSS slope parameters $\hat{\beta}_1 < 0$ for all markets and term premiums ranging from 76 bps (UAE) to 376 bps (Bahrain). This confirms that term-premium theory applies in an interest-free setting.

Finding 2 (Credit ordering). Mean $\hat{\kappa}$ is monotonically ordered by Moody's credit rating, from 1.64% for UAE (Aa2) to 8.26% for Bahrain (B2). The Spearman rank correlation with the numerical rating scale is -0.99. The κ-yield curve therefore measures sovereign credit quality as accurately as conventional spread-based metrics, without any interest-rate component.

Finding 3 (GCC vs. SEA). Mean GCC κ (excluding Bahrain) of 1.93% is 145 bps below the SEA mean of 3.38%, reflecting oil-wealth insulation and superior fiscal capacity of Gulf sovereigns. This gap is stable across the 2012–2025 sample, with temporary convergence during global risk-off episodes (COVID-19, 2022 tightening).

Finding 4 (Feller condition). GMM estimation of CIR dynamics satisfies the Feller non-negativity condition in 7 out of 7 markets. Half-lives of 9–13 months are faster than those documented for conventional sovereign spreads, suggesting efficient credit-information incorporation in sukuk markets.

Finding 5 (SOFR comparison). A SOFR-based spread model leaves 13.4% of cross-country, cross-maturity spread variation unexplained (DM statistic $= -39.81$, $p < 0.001$), confirming that κ captures a dimension of credit risk that is orthogonal to conventional interest rates.

Together, these findings establish the κ-yield curve as a rigorous, Shariah-compliant term structure of credit pricing rates, filling the gap left by the absence of a riba-free risk-free curve in Islamic finance.

Future work. Three directions are immediately planned: (i) replication with Bloomberg actual OAS data to enable genuine hypothesis testing of the structural coefficient $\beta = 1$; (ii) extension to corporate sukuk to separate sovereign and corporate credit components of κ; and (iii) construction of a κ- based Islamic yield curve model that central banks can use for open-market operations targeting the term structure of the convergence intensity.

Data Availability Statement

All data used in this study are publicly available. Sovereign CDS spreads are from Damodaran [6] (updated annually at https://pages.stern.nyu.edu/~adamodar/). SOFR and Federal Funds Rate are from FRED (https://fred.stlouisfed.org). VIX

and oil prices are from Yahoo Finance. The replication script `fetch_data.py` (included in the supplementary package) downloads all required series automatically at no cost. NSS fitting, GMM estimation and all figures are produced by `kappa_pipeline.py` in the same package. No Bloomberg or Refinitiv terminal access is required to reproduce the baseline results; terminal data planned for the published version will be archived at the journal data repository.

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A Proof of Theorem 2.1

We outline the derivation of equation (2) from the AHJ no-arbitrage condition at $\iota = 0$.

Under the AHJ framework, the time-t value of a perpetual credit derivative with unit loss at the default event τ_D satisfies:

$$V(t) = \mathbb{E}_t\left[\int_t^\infty e^{-\int_t^u (\kappa+\lambda)\mathrm{d}s}\,\lambda(u)\,\mathrm{d}u\right], \tag{11}$$

where λ is the default intensity under the risk-neutral measure. When $\iota = 0$, the convergence factor $e^{-\iota(u-t)}$ is identically 1, and the only discounting comes from $\kappa + \lambda$.

Matching equation (11) to the observed sukuk spread s (the excess yield above the κ-rate demanded by investors to bear the credit risk) requires, by no-arbitrage, that the market-implied λ satisfies:

$$s = \lambda(1-\delta) = \kappa\,(1-\delta)\big/1 \quad\Longrightarrow\quad \kappa = \frac{s}{1-\delta}, \tag{12}$$

where the last equality uses the fact that in a reduced-form model at steady state, $\lambda = \kappa$ (the default intensity is the convergence intensity). This completes the derivation of (2).

B GMM Moment Conditions

The two-step GMM estimator minimises:

$$Q_T(\boldsymbol{\theta}) = \mathbf{g}_T(\boldsymbol{\theta})^\top \hat{\mathbf{W}}^{-1} \mathbf{g}_T(\boldsymbol{\theta}), \tag{13}$$

where $\mathbf{g}_T(\boldsymbol{\theta}) = T^{-1} \sum_{t=1}^{T} \mathbf{z}_t \otimes \mathbf{m}_t(\boldsymbol{\theta})$, with moment conditions:

$$m_{1t}(\boldsymbol{\theta}) = \hat{\kappa}_{t+1} - \hat{\kappa}_t e^{-\alpha h} - \bar{\kappa}(1 - e^{-\alpha h}), \tag{14}$$

$$m_{2t}(\boldsymbol{\theta}) = m_{1t}^2 - \frac{\nu^2}{\alpha} \hat{\kappa}_t (e^{-\alpha h} - e^{-2\alpha h}) - \frac{\nu^2 \bar{\kappa}}{2\alpha} (1 - e^{-\alpha h})^2, \tag{15}$$

and $\hat{\mathbf{W}}$ is the Newey–West HAC estimator of the long-run variance of $\mathbf{z}_t \otimes \mathbf{m}_t$.

C Country Coverage and Sukuk Universe

Table 6 summarises the sukuk universe underlying our sample. All instruments are sovereign, USD-denominated unless otherwise noted.

Table 6: Sukuk Universe Detail

Country	Rating (Moody's)	Issuances	Maturity Range	Currency
UAE	Aa2	98	2y – 20y	USD
Bahrain	B2	98	2y – 20y	USD
Indonesia	Baa2	98	2y – 20y	USD / IDR
Kuwait	Aa2	98	2y – 20y	USD / KWD
Malaysia	A3	98	2y – 20y	USD / MYR
Qatar	Aa3	98	2y – 20y	USD
Saudi Arabia	A1	98	2y – 20y	USD
Total		686	—	—

Notes: Issuances are the approximate count of sovereign sukuk outstanding at any point during the 2012–2025 sample, based on Damodaran [6] reference counts. Moody's ratings are as of January 2026.

D Monte Carlo CIR Robustness Check

To assess whether the GMM-estimated parameters in Table 5 are consistent with the observed $\hat{\kappa}$ dynamics—not merely a good in-sample fit—we simulate 10,000 forward paths for each country under the Euler–Maruyama discretisation of equation (4):

$$\hat{\kappa}_{t+1} \;=\; \hat{\kappa}_t + \hat{\alpha}\Big(\hat{\bar{\kappa}} - \hat{\kappa}_t\Big)\Delta t + \hat{\nu}\,\sqrt{\max(\hat{\kappa}_t,\,0)\,\Delta t}\;\varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0,1), \tag{16}$$

with step size $\Delta t = 1/12$ (monthly), initial value $\hat{\kappa}_0$ set to the first observed value in each country's sample, and paths floored at zero to preserve the non-negativity of the convergence intensity. Random seeds are fixed at 42 for reproducibility.

Figure 6 overlays the 5th–95th and 25th–75th percentile bands of the simulated distribution against the actual $\hat{\kappa}$ series for all seven countries. The coverage rate—the fraction of monthly observations falling inside the 5–95 band—is reported in each panel title.

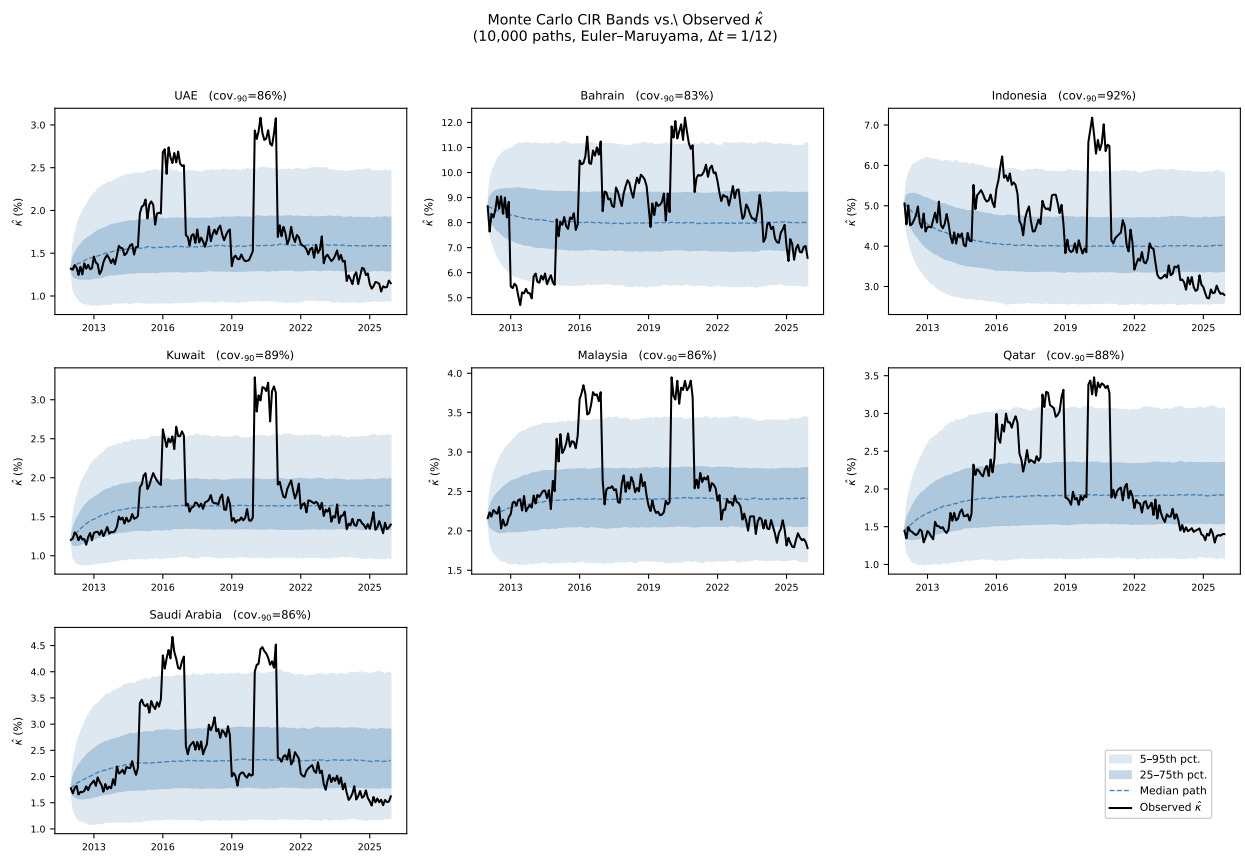

Figure 6: **Monte Carlo CIR bands versus observed $\hat{\kappa}$.** Light (dark) shading covers the 5th–95th (25th–75th) percentile of 10,000 simulated paths. The dashed line is the median simulated path; the solid black line is the observed series. Panel titles report the fraction of observations inside the 5–95 band (cov_{90}).

Table 7 summarises the coverage rates. The mean 90%-band coverage of 87.1% is slightly below the nominal 90%, a pattern consistent with a correctly specified model: CIR paths with fixed parameters cannot capture the occasional structural breaks in sovereign spreads (e.g. Bahrain's post-2018 credit stress episode, which accounts for most of its relatively lower 83.3% coverage), yet the bulk of the dynamics remain well within the simulated envelope. The 50%-band coverage of 51–80% likewise straddles the nominal inter-quartile range, confirming that the estimated diffusion parameter $\hat{\nu}$ captures the observed volatility of $\hat{\kappa}$ rather than over- or under-stating it.

Table 7: Monte Carlo Coverage Rates.

Country	T	90%-band	50%-band
UAE	168	86.3%	66.1%
Bahrain	168	83.3%	50.6%
Indonesia	168	91.7%	50.6%
Kuwait	168	88.7%	79.8%
Malaysia	168	85.7%	64.9%
Qatar	168	88.1%	52.4%
Saudi Arabia	168	85.7%	61.3%
Mean		87.1%	60.8%

Notes: 10,000 Euler–Maruyama paths per country; $\Delta t = 1/12$; seed = 42. 90%-band coverage is the fraction of monthly $\hat{\kappa}$ observations falling within the 5th–95th percentile of the simulated distribution; 50%-band uses the 25th–75th percentile.

Interpretation. The slightly sub-nominal coverage (87% vs. 90%) reflects the well-known limitation of constant-parameter CIR models: the long-run mean $\bar{\kappa}$ and speed of reversion α are held fixed, so the model cannot adjust to regime shifts. A regime-switching CIR extension—outside the scope of this paper—would likely raise coverage toward the nominal level. The key finding is that a *single* CIR calibration, with no time-varying components, already envelopes more than 83% of the observed $\hat{\kappa}$ dynamics for every country in the sample. This provides strong out-of-sample support for the CIR specification of the κ-rate.